THRIFTKART / SEMESTER III

ThriftKart Project Master

One-page-to-full-baseline project blueprint

Project	Second-Hand Textbook Marketplace for College Students
Course	Software Engineering & Project Management
Case	No. 72
Core artefacts	BRD • SRS • UML • Project Plan • Test Plan • Risk Register
Product	Campus-first resale web application
Delivery stance	Finishable MVP + polished frontend prototype

1. What This Project Is

ThriftKart is a web marketplace designed for college students to resell used textbooks to other students. Seniors become sellers; juniors become buyers; the transaction is completed on campus. The product exists to reduce the cost of short-life textbook purchases while creating a structured, moderated alternative to informal resale channels.

2. What the Case Study Expects

The supplied case asks for requirements engineering, an internally consistent UML package, measurable NFRs, an MVP decision, three-point estimation, a schedule with critical path and float, systematic testing including boundary-value analysis and decision tables, defect density/DRE, and a risk register with RMMM. The case explicitly calls for an IEEE 830-style SRS, UML package, project plan, estimation sheet and test evidence.

₹6,500	₹1.5L	9 weeks	4 × 3h
case-given annual textbook spend	incubation grant	exam constraint context	weekday team capacity

3. Product Vision

Vision

Make the next student's textbook cheaper, easier to find and safer to buy — without adding delivery or payment infrastructure that the team cannot finish before exams.

4. Product Modules

Module	What it does
Access	Registration, login, role-aware experience.
Marketplace	Search, filters, listing cards, detail view.
Selling	Create draft, validate, submit for moderation.
Trust	Moderation, reports, decision evidence.
Conversation	Listing-specific lightweight buyer/seller chat.
Transaction	Purchase request, payment status, campus handover, completion.
Admin	Moderation queue and report handling.

5. MVP Decision

Area	MVP choice	Why
Chat	Lightweight in-app chat	Useful for trust and handover coordination; can stay simple and listing-scoped.
Payments	Status + confirmation, not gateway	Preserves the sale workflow without adding external payment complexity.
Delivery	Not included	Campus handover is the case's fulfillment model.
Recommendations	Not included	Does not contribute to the primary case outcome.
Advanced notifications	Not included	In-app transaction state is sufficient for the baseline.

6. Database / Data Model

Table	Core fields	Keys / constraints	Role in product
users	user_id, university_email, password_hash, full_name, year_of_study, role, status, created_at	PK user_id; UNIQUE university_email	Student/admin identity and access
categories	category_id, name, slug	PK category_id; UNIQUE slug	Textbook grouping
listings	listing_id, seller_id, category_id, title, author, isbn, edition, course_code, price, condition_rating, description, campus_location, status, created_at, updated_at	PK listing_id; FK seller_id -> users; FK category_id -> categories	Core textbook marketplace record
listing_images	image_id, listing_id, image_url, sort_order	PK image_id; FK listing_id -> listings	One-to-many listing media
conversations	conversation_id, listing_id, buyer_id, seller_id, status, created_at	PK conversation_id; FKs to listings/users; UNIQUE(listing_id,buyer_id,seller_id)	Buyer-seller conversation per listing
messages	message_id, conversation_id, sender_id, body, sent_at, read_at	PK message_id; FK conversation_id -> conversations; FK sender_id -> users	Chat messages
transactions	transaction_id, listing_id, buyer_id, seller_id, agreed_price, payment_method, payment_status, handover_location, handover_at, status, created_at, completed_at	PK transaction_id; FKs to listings/users	Sale lifecycle and campus handover
payment_records	payment_id, transaction_id, amount, method, provider_reference, status, paid_at	PK payment_id; FK transaction_id -> transactions	Payment confirmation/audit record; gateway optional
moderation_reviews	review_id, listing_id, moderator_id, decision, reason, reviewed_at	PK review_id; FK listing_id -> listings; FK moderator_id -> users	Approval/rejection evidence
reports	report_id, reporter_id, listing_id, reported_user_id, reason, status, created_at, resolved_at	PK report_id; FKs to users/listings	Trust and safety reporting queue

7. Requirements Snapshot

Functional requirement	Priority	Key screen/module
FR-01	Must	Account access
FR-02	Should	Profile management
FR-03	Must	Create listing
FR-04	Must	Listing validation
FR-05	Must	Moderation workflow
FR-06	Must	Marketplace search
FR-07	Must	Listing detail
FR-08	Should	Buyer-seller chat
FR-09	Must	Purchase request
FR-10	Must	Payment status
FR-11	Must	Campus handover
FR-12	Should	Trust and safety

Use the SRS as the authoritative requirement wording. The full set is 12 FRs and 8 NFRs, with every FR mapped to a named test case.

8. Three-Point Estimation

PERT expected time formula: TE = (O + 4M + P) / 6.

Work package	O	M	P	TE (days)

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Listing & search	4	6	12	6.67
Chat between buyer and seller	3	5	10	5.5
Payments	3	4	9	4.67
Admin moderation	2	3	6	3.33
Testing	3	4	7	4.33

Total effort

6.67 + 5.50 + 4.67 + 3.33 + 4.33 = 24.50 productive days. Team capacity = 4 × 3 = 12 hours/day. At 6 productive hours = 1 day, capacity = 2 productive days/day = 10 productive days/week. A purely serial conversion is therefore 24.50 / 10 = 2.45 calendar weeks.

9. Schedule, Forward Pass & Critical Path

A practical dependency network is: Listing/Search → Chat → Payments → Testing, with Admin Moderation running in parallel after Listing/Search and also feeding Testing. This creates a non-critical moderation branch, which lets the project demonstrate float as required by the case.

ID	Task	Dur.	ES	EF	Float	Path
A	Listing & search	6.67	0.00	6.67	0.00	Critical
B	Chat	5.50	6.67	12.17	0.00	Critical
C	Payments	4.67	12.17	16.84	0.00	Critical
D	Admin moderation	3.33	6.67	10.00	6.84	Non-critical
E	Testing	4.33	16.84	21.17	0.00	Critical

Critical path

Listing & search → Chat → Payments → Testing = 21.17 productive days ≈ 2.12 weeks of critical-path effort. Admin moderation has 6.84 days of float under this dependency model. The 24.50-day figure is total estimated effort, not a promise of calendar completion.

10. Four Milestones

Milestone	Name	Checkpoint meaning
M1	Requirements & scope baseline	Checkpoint after requirements / MVP decisions are locked.
M2	Design baseline	UML + data model + architecture internally consistent.
M3	Feature complete	Core marketplace, moderation and transaction workflow available for system testing.
M4	Assessment release	Testing evidence, metrics, risk closure note and final documents assembled.

11. Test Strategy

Technique	What to test	Expected evidence
Boundary Value Analysis	Price and condition rating	Min/max valid values + just-outside invalid values.

Technique	What to test	Expected evidence
Equivalence Classes	Valid/invalid price and condition groups	Representative test cases for each class.
Decision Table	Moderation and price rules	All meaningful combinations with expected approve/reject result.
Integration tests	Listing → moderation → request → transaction	State transitions and data integrity.
Defect metrics	Defect log	Defect density and DRE computed from recorded results.

12. Boundary / Decision Table Seeds

Rule	Test values / conditions	Expected
Price range	Proposed bounds ₹1–₹20,000; test 0, 1, 20,000, 20,001	0/20,001 reject; 1/20,000 accept
Condition	Proposed integer range 1–5; test 0, 1, 5, 6	0/6 reject; 1/5 accept
Moderation	Required fields complete? Content acceptable? Duplicate/suspicious?	Approve only when all required controls pass
Reservation	Listing available? Existing active reservation?	Create reservation only when available and unreserved

13. Risk Register & RMMM

ID	Risk	Prob.	Impact	RMMM	Owner / control
R1	Scope creep	High	High	Freeze MVP; defer gateway/delivery/recommendations.	Team lead + mentor review
R2	Exam-week capacity loss	High	High	Front-load critical path; keep exam weeks at zero work.	Weekly plan review
R3	Trust / fake listings	Medium	High	Moderation gate, reporting, evidence log.	Admin moderator
R4	Payment complexity	Medium	Medium	Use status-based payment confirmation in MVP.	Backend owner
R5	Inconsistent artefacts	Medium	High	Trace FRs to tests and keep UML/data model synchronized.	BA / QA

14. What You Should Draw for UML

- Use Case Diagram: Student Buyer, Student Seller, Admin Moderator; optionally a generic Student actor generalized into buyer/seller.
- Class Diagram: User, Category, Listing, ListingImage, Conversation, Message, Transaction, PaymentRecord, ModerationReview, Report; show multiplicities and key methods/state attributes.
- Sequence Diagram: List a Book and Complete a Sale, showing Seller, Web UI, API, Database, Admin, Buyer, Chat/Transaction modules.
- Activity Diagram: branch for seller listing submission and buyer purchase/handover flow.
- State Diagram: Draft → Pending Moderation → Published → Reserved → Sold; alternative Rejected and Archived transitions.

Cohesion & coupling justification

Keep Listing/Marketplace responsible for book records and discovery; keep PaymentRecord responsible for payment evidence/status; keep Transaction responsible for the sale lifecycle. Modules communicate through IDs and service boundaries rather than embedding payment logic inside listing logic. This is high cohesion with loose coupling.

15. Final Deliverable Checklist

Artefact	Ready state
BRD	Business problem, objectives, stakeholders, process, scope, rules, success criteria, approval table.
SRS	12 FRs, 8 measurable NFRs, interfaces, data model, use cases, MoSCoW, traceability.
UML package	Use case, class, sequence, activity and state diagrams, all internally consistent.
Project plan	WBS, dependency network, forward pass, critical path, float, milestones, resource allocation.
Estimation sheet	All O/M/P values and TE formula + capacity conversion.
Test evidence	BVA, equivalence classes, decision table, defect log, DRE, defect density.
Risk register	Probability-impact, exposure-ranked risks, RMMM for top risks, closure note.
Website prototype	Polished marketplace UI reflecting the same data/flows as the documents.

16. 30-Second Viva Explanation

Say this

“ThriftKart is a campus-first second-hand textbook marketplace. Sellers list books, listings pass through moderation, buyers search and request a book, payment status is recorded, and the sale finishes through campus handover. The MVP is deliberately small so the four-student team can complete it before the exam constraint, while the SRS, UML, estimates, tests and risk register provide engineering evidence.”